

2016 APMCM Problem A

Temperature and key element content prediction based on optical information data

Light contains energy, and it can be converted into heat in some certain conditions. Burning is an example as common phenomenon which can give both light and heat. Light and heat are usually concomitant, and generally speaking, the higher light intensity, the higher the temperature.

There is no direct relationship between light intensity and its temperature, but the optical frequency has a great relationship with the temperature, such as low frequency infrared light and far infrared light, which has a typical thermal effect, while there is nearly no thermal effect for UV light with high frequency. In the case of the same frequency, the larger the light intensity, the better the thermal effect.

Metal smelting is a process that metal material thrown into a given melting pot will be smelt under a certain temperature in order to control or eliminate inferior element and remain or increase the proportion of superior element. Thus seek to acquire the best performance of objective metal. So the key factor of metal smelting lies in the control of temperature and the content of key elements. In the process of metal smelting, the furnace produces flame, then the optical information emitted by the flame is projected onto the photo detector through the theory of pinhole imaging. Next, according to the discrete frequency, the light intensity data of the flame is recorded at every 0.5s by the photo detector. Optical information data generation process at a certain time:

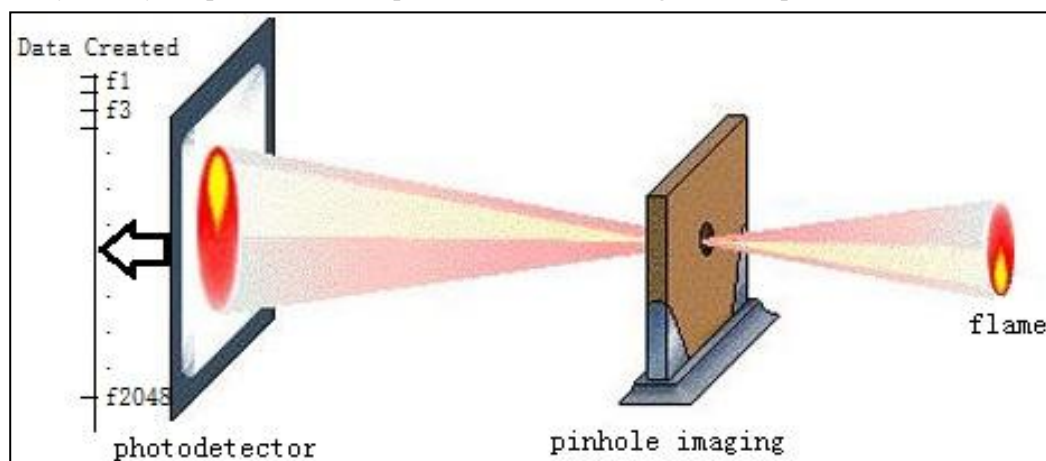


Fig. Optical data generation process at a certain time

From the image above, the amount of data received by the photo detector is 2048 during every 0.5 seconds. It is required to use the values of these 2048 light intensity to predict the flame temperature and the content of the key elements in the raw materials in real time.

The data in the three process of metal smelting are presented in the appendix, which are 1.xlsx,

2.xlsx and 3.xlsx. The data includes the time t , the cumulative consumption of combustion gas Q , the cumulative consumption ratio of combustion gas p , optical information data (f_1 - f_{2048} , light intensity at different frequencies), Kelvin temperature T and content of key element C .

Glossary:

Optical information characteristics: If considering 2048 light intensity data generated at a time as the input and the flame temperature and the Key element content as the output, constructing a mathematical model is bound to be complex, so that it can not achieve the forecast results. In order to reduce the number of input data, try to find one or more characteristic values of light intensity data, which will be used as input to reduce the complexity of the model calculation, and enhance the applicability of the model.

Crossover experiment: In the title, the data in the process of three metal smelting is given, and the prediction model of temperature and the Key element content was established based on three different processes. In order to verify whether the model is universal, the model of the 1 process is simulated in the 2 process and 3 process. And then explore the error characteristics. Crossover experimental design is shown in the following table:

Table Crossover experiment design

Data Model	Prediction model based on 1 process	Prediction model based on 2 process	Prediction model based on 3 process
1 process data	Err11	Err12	Err13
2 process data	Err21	Err22	Err23
3 process data	Err31	Err32	Err33

Err11 represents the self inspection error based on the prediction model of 1 process, Err21 represents the crossover inspection error based on the prediction model of 1 process and the data of 2 process, and so on.

Through the above background, explore the following three questions:

1) The mathematical model or algorithm model should be established by the data given in the annex to find the characteristic λ of the optical information data and extract the features ($\lambda_1, \lambda_2, \dots, \lambda_n$) into the Excel table. The data files name are: feature_output_1.xlsx, feature_output_2.xlsx and feature_output_3.xlsx.

2) The mathematical model or algorithm model should be established by using the optical information characteristic data extracted in 1) and the data given in the data table, such as the time t and the accumulated consumption Q of the combustion- supporting gas, to predict the Kelvin temperature T and key element content C and to explore the relationship.

3) Through the exploration of question 1) and question 2), design the crossover experiment

scheme, cross-validate the error generated by the prediction target, and provide the error control scheme on the basis of the error analysis.